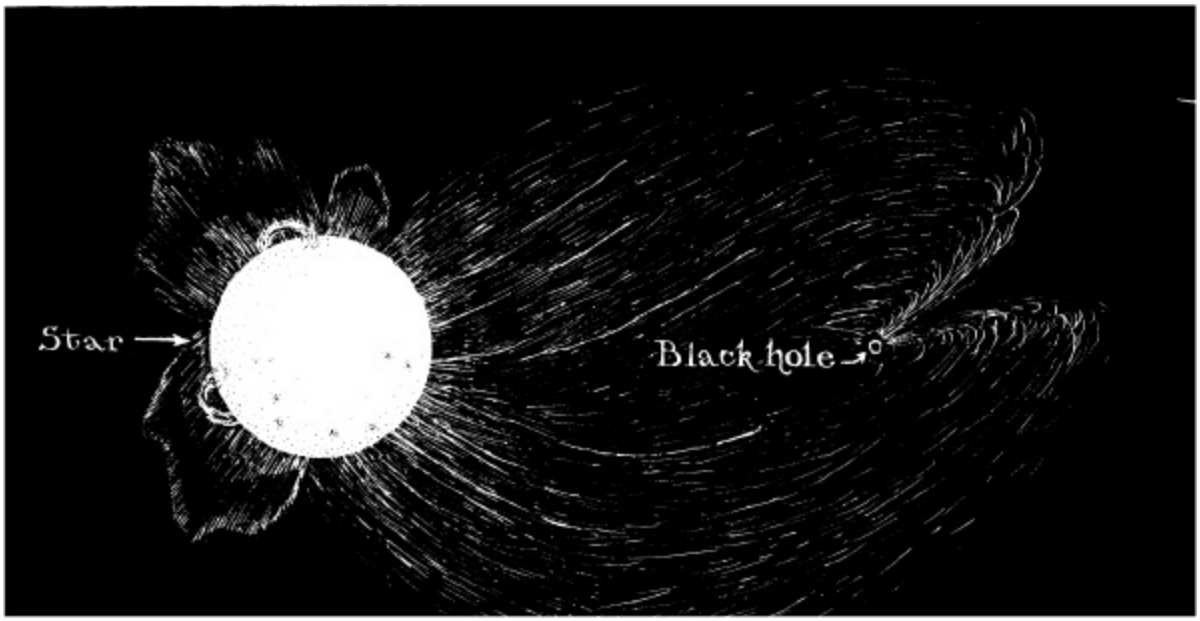
Zel’dovich and his team mulled over this idea for two years, looking at it first from this direction and then that, searching for ways to make it more promising. However, it was but one of dozens of ideas about black holes, neutron stars, supernovae, and the origin of the Universe that they were pursuing, and it got only a little attention. Then, one day in 1966, in an intense discussion, Zel’dovich and Novikov together realized they could combine the binary star idea with the infalling gas idea (Figure 8.5).

Strong winds of gas (mostly hydrogen and helium) blow off the surfaces of some stars. (The Sun emits such a wind, though only a weak one.) Suppose that a black hole and a wind-emitting star are in orbit around each other. The hole will capture some of the wind’s gas, heat it in a shock front, and force it to radiate. At the one-meter-square black-board in Zel’dovich’s Moscow apartment, he and Novikov estimated the temperature of the shocked gas: several million degrees.



8.5 The Zel’dovich-Novikov proposal of how to search for a black hole. A wind, blowing off the surface of a companion star, is captured by the hole’s gravity. The wind’s streams of gas swing around the hole in opposite directions and collide in a sharp shock front, where they are heated to millions of degrees temperature and emit X-rays. Optical telescopes should see the star orbiting around a heavy, dark companion. X-ray telescopes should see X-rays from the companion.

Gas at such a temperature does not emit much light. It emits X-rays instead. Thus, Zel’dovich and Novikov realized, among those black holes